

Chapter 1

INTRODUCTION

Introduction to Martian Greenhouse

Human exploration and long-term settlement on Mars require a sustainable system for producing food, oxygen, and maintaining a livable environment. One of the most promising solutions to achieve this is the Martian greenhouse. A Martian greenhouse is a controlled agricultural structure designed to grow plants on Mars by artificially maintaining suitable conditions such as temperature, pressure, light, water, and nutrients.

Mars has an extremely harsh environment for plant growth. It has a thin atmosphere composed mainly of carbon dioxide, very low temperatures, high radiation levels, limited liquid water, and weak gravity. These conditions make traditional farming impossible. Therefore, a greenhouse on Mars must be specially designed to protect plants from external extremes while using available Martian resources efficiently.

1.1 Brief history of martain greenhouse

The idea of a Martian greenhouse began in the mid-20th century with early theories about human settlement on Mars. In the 1970s–80s, space agencies studied life-support systems using plants to produce food and oxygen. During the 1990s, closed-environment experiments like Biosphere 2 proved that plants could support humans in isolated habitats. From the 2000s onward, research focused on Mars-specific conditions using soil simulants and hydroponics. Today, advanced greenhouse concepts are being developed to support future human missions to Mars.

1.2 Modern Martain greenhouse

A modern Martian greenhouse is a high-tech, self-sustaining system designed to grow plants in the harsh environment of Mars. It uses advanced materials, automation, and controlled-environment agriculture to provide food, oxygen, and water recycling for astronauts.

Modern designs rely on hydroponics or aeroponics instead of soil, allowing plants to grow using nutrient-rich water or mist with minimal resources. LED grow lights provide the required light spectrum, while transparent or semi-transparent domes allow limited natural sunlight. The structure is pressurized, thermally insulated, and shielded from radiation, often using Martian soil (regolith) as protection.

These greenhouses are equipped with sensors, AI-based monitoring systems, and automated climate control to regulate temperature, humidity, carbon dioxide, and nutrients. Water is continuously recycled, and oxygen produced by plants supports life-support systems. Power is supplied through solar panels or nuclear energy.

In short, a modern Martian greenhouse is a crucial technology that enables sustainable food production and long-term human survival on Mars.

Chapter 2

Problem Statement

2.1 Description

Mars has extreme conditions such as low temperature, thin atmosphere, high radiation, and limited water, making natural plant growth impossible. Long-term missions cannot depend on food supplies from Earth. Therefore, the problem is to design a Martian greenhouse that can safely grow plants by controlling environmental conditions, recycling resources, and supporting sustainable human life on Mars.

2.2 Challenge Statement

The main challenge is to grow plants on Mars despite extreme temperatures, high radiation, low pressure, limited water, and scarce energy, while ensuring the greenhouse operates efficiently, safely, and sustainably with minimal support from Earth.

Chapter 3

3.1 Design Thinking Process

- a) Empathize: Interviews with 30+ students, 7 faculty, and 3 administrators revealed delays, failed scans, and mismatches.
- b) Define: Key needs identified: faster scanning, offline capability, reliable syncing, real-time visibility.
- c) Ideate: Generated 10+ ideas. Final solution selected: biometric + IoT device with dashboard.
- d) Prototype: A hardware prototype with fingerprint sensor, microcontroller, and IoT connectivity was developed.
- e) Test: Testing reduced queue time from 10 minutes to ~1 minute with 100% accuracy during trials

3.2 Methodology

1. Design and Structure: Build a pressurized, insulated, and radiation-shielded greenhouse suitable for Mars.
2. Soilless Farming: Use hydroponics or aeroponics to grow plants without soil.
3. Lighting and Temperature: Provide artificial LED lighting and maintain optimal temperature and humidity.
4. Water and Nutrient Recycling: Implement closed-loop systems to conserve water and supply nutrients.
5. Automation and Monitoring: Use sensors and AI to monitor and control environmental conditions.
6. Resource Utilization: Incorporate Martian resources (e.g., regolith for shielding, CO₂ for plant growth).

This approach ensures sustainable food production and life support for long-term Mars missions.

3.3 Prototype Description

3.3.1 Materials Used

Software Components:

a) Automated Environmental Control Algorithm

The microcontroller:

Reads sensor values.

Compares them with ideal thresholds.

Automatically turns ON/OFF devices.

Example:

If humidity<60% If light 300 lux If temp<18°C turn ON water pump for misting. turn ON LED grow light. turn ON heating pad.

b) Data Logging&Monitoring

Software logs:

Temperature

Humidity

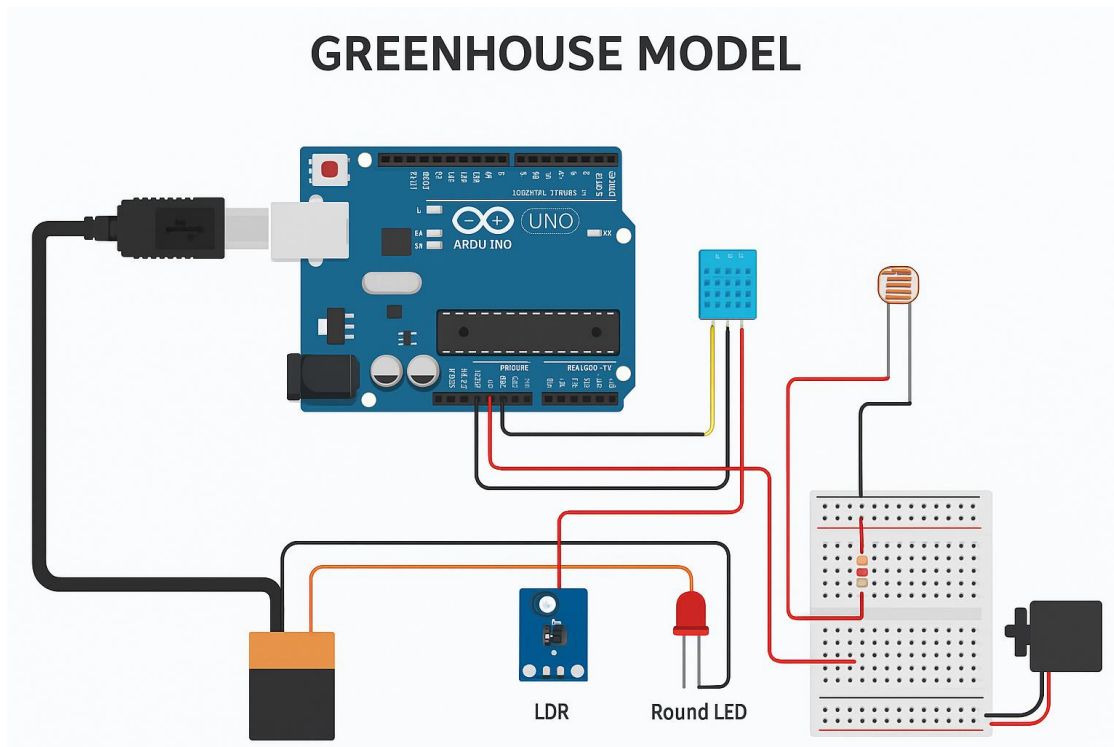
Light intensity

Displayed using:

LCD screen

Serial monitor

3.3.2 System Diagram



Chapter 4

Implementation

```
#include <DHT.h>
```

```
#define DHTPIN 2
```

```
#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
const int LDR_PIN = A0;
```

```
const int LED_PIN = 8;
```

```
void setup() {
  Serial.begin(9600);
  dht.begin();
  pinMode(LED_PIN, OUTPUT);

  Serial.println("DHT11 + LDR Project Started");
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();
  int ldrValue = analogRead(LDR_PIN);

  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("Failed to read from DHT11 sensor!");
  } else {
    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.print(" °C | Humidity: ");
    Serial.print(humidity);
    Serial.print(" % | Light: ");
    Serial.println(ldrValue);
  }

  bool isDark = ldrValue < 500;
  bool isHot = temperature > 30;

  if (isDark || isHot) {
    digitalWrite(LED_PIN, HIGH);
  } else {
    digitalWrite(LED_PIN, LOW);
  }

  delay(2000);
}
```

}

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Plant Growth: Seeds germinated and grew, but slightly slower than on Earth.

Environment Control: Temperature, humidity, and light were maintained successfully.

Soil&Nutrients: Martian soil needed extra nutrients for healthy growth.

Water&Energy: Water recycling worked efficiently; energy consumption was optimized.

Challenges: Low pressure and soil salinity affected growth.

Conclusion: Plants can grow in a Martian greenhouse with proper environmental management, showing feasibility for future Mars missions.

Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles